

Technology Stack Review

An independent assessment of every component in the v1 stack -- whether it remains the correct selection given what is currently available, what the alternatives are, and the recommendation for each.

Overall finding: Every component in the current stack is the correct selection for its role. No swaps are recommended. The stack is appropriately lean for a solo developer operating under UK data residency, a GBP100/month cost ceiling, and a 31 July 2026 launch target. One addition -- Axiom for structured log management -- is worth scheduling for P5.4 to close the AI route latency observability gap.

Component Review

TS-01 Framework & Language		Retain
Current:	Next.js App Router + TypeScript	
Alternatives:	Remix / React Router v7 -- Merged with React Router in late 2024; strong form handling. Smaller ecosystem; no Vercel parity advantage; fewer mature AI/SaaS library integrations. SvelteKit -- Lighter runtime, excellent DX. Smaller future-maintainer pool -- a real concern given the C18 succession plan -- and fewer WCAG 2.2 AA component libraries.	
Recommendation:	<i>Next.js App Router is the industry standard for this class of product. The Vercel-native deployment, breadth of ecosystem, and the largest future-maintainer pool are genuine advantages -- not defaults.</i>	

TS-02 Database		Retain
Current:	Supabase PostgreSQL -- London region	
Alternatives:	Neon -- Serverless Postgres with database branching. Lacks Auth and Storage; would require two additional vendors. No equivalent to Supabase native RLS-auth integration. PlanetScale -- Deprecated its free tier in 2024. MySQL lacks PostgreSQL features in use here. No London region. Turso (libSQL) -- Designed for edge-distributed SQLite reads; not a natural fit for complex relational queries under strict UK data residency.	
Recommendation:	<i>The combination of Postgres, Auth, RLS, and Storage in a single London-region service is unique in the market. No alternative bundles these capabilities without introducing additional vendors.</i>	

TS-03 Authentication		Retain
Current:	Supabase Auth (bundled)	
Alternatives:	Clerk -- Excellent DX; pre-built React components. UK/EU data residency requires the Enterprise plan. Removes the native RLS-auth linkage that automatically scopes every database query to the authenticated user. Auth.js (NextAuth v5) -- Open-source, no vendor lock-in. Requires custom RLS integration logic and significantly more session-management plumbing for a solo developer. Lucia -- Lightweight, session-based, good control. Requires writing more auth plumbing code at the cost of development velocity.	
Recommendation:	<i>Supabase Auth's native RLS integration is a genuine architectural advantage: every user session automatically scopes database access at the database level. Clerk or Auth.js cannot replicate this without custom work.</i>	

TS-04 Hosting Platform		Retain
Current:	Vercel Pro -- London region (lhr1)	
Alternatives:	Cloudflare Pages + Workers -- Edge-first, globally fast, significantly cheaper. Workers have a CPU time cap that cuts AI routes well below the 30-45 s call times observed in testing. The 90-second timeout requirement cannot be met without a full Durable Objects redesign. Fly.io -- Container-based, London region available. More operational overhead; no first-class Next.js optimisation; no equivalent per-PR preview deployments. Railway -- Good developer experience. No CDN equivalent; no Next.js-specific optimisations; preview deployments require additional configuration.	
Recommendation:	<i>The 90-second function timeout eliminates Cloudflare -- the only serious cost-competitive alternative. Vercel with a London function region aligns compute directly with Bedrock eu-west-2, keeping all processing in UK infrastructure.</i>	
TS-05 Email Service		Retain -- Monitor
Current:	Resend (free tier)	
Alternatives:	AWS SES (eu-west-2) -- Already have active AWS credentials in the same region. Cost ~GBP0.10 per 1,000 emails. Would consolidate billing and reduce vendor count by one. Requires SMTP + DKIM configuration in Supabase Auth settings. Postmark -- Best-in-class transactional deliverability. More expensive; no meaningful advantage at current volumes (under 200 emails/month). Brevo -- EU-headquartered, GDPR-friendly, 300 emails/day free. Less developer-focused API; Resend DX is cleaner for a code-first setup.	
Recommendation:	<i>Retain now. At current volumes the Resend free tier is more than sufficient. When email volume becomes a concern, migrating to AWS SES is a clean swap -- Supabase SMTP config change plus package removal -- that consolidates billing within the existing AWS footprint.</i>	
TS-06 Error Tracking		Retain
Current:	Sentry EU	
Alternatives:	Highlight.io -- Open-source; session replay included; EU cloud and self-hosting available. Less mature for Next.js edge runtime coverage. Bugsnag -- Reliable, used by large teams. More expensive; less Next.js-specific SDK maturity than Sentry. Vercel logs only -- No persistent history, no deduplication, no alerting. Sufficient for development; not for production monitoring of a live charity service.	
Recommendation:	<i>Sentry's Next.js SDK covers all three runtimes (client, server, edge) and the EU data region satisfies C13. Complement with Axiom at P5.4 for structured log management -- see New Addition section.</i>	
TS-07 Rate Limiting		Retain
Current:	Upstash Redis -- sliding window	
Alternatives:	Supabase DB counter -- Eliminates Upstash as a separate vendor but adds database load on every AI request and provides weaker burst protection than an atomic Redis counter. Vercel KV -- Upstash Redis under the hood. Identical capability; would add a Vercel billing dependency without reducing vendor count. Application-level only -- The monthly ai_usage_log cap blocks sustained abuse but provides no protection against rapid-fire bursting within the monthly limit.	

Recommendation:	<i>The sliding window algorithm provides measurably better burst protection than a DB count, at zero additional cost -- the Upstash free tier covers 200,000 requests per day, far above expected AI route volume.</i>
TS-08 CI Pipeline	
Current:	GitHub Actions -- 4 jobs
Alternatives:	CircleCI -- Mature CI platform. No advantage over GitHub Actions for a public GitHub repository; introduces an additional account and configuration surface. GitLab CI -- Would require migrating the repository away from GitHub. No benefit; significant disruption.
Recommendation:	<i>No credible alternative offers any advantage for a public GitHub repository. GitHub Actions is zero-configuration, free for public repos, and covers lint, type-check, tests, security audit, and migration validation on every push.</i>
TS-09 Test Framework	
Current:	Vitest
Alternatives:	Jest -- Slower than Vitest in CI; TypeScript requires additional configuration (ts-jest or Babel). No meaningful advantage for a TypeScript-first project. Playwright -- End-to-end browser testing; complementary to Vitest rather than a replacement. Worth adding post-launch when UI flows are stable.
Recommendation:	<i>Vitest is the correct choice: Jest-compatible API, native TypeScript support, significantly faster in CI. Playwright is the natural next addition when E2E coverage becomes a priority after launch.</i>
AI AI Provider & Model	
Current:	Amazon Bedrock eu-west-2 -- Claude Sonnet 4.6
Alternatives:	Azure AI Foundry (Claude) -- Azure now offers Anthropic Claude models in EU regions including UK South. SDK change required; no quality or cost advantage -- trades one cloud dependency for another. Anthropic direct API -- US-region only as of mid-2025; no EU endpoint available. Cannot satisfy C13 data residency requirement. Claude Opus 4.8 -- Higher quality, significantly higher cost. Output quality with Sonnet 4.6 is already strong for grant writing; Opus is not justified at v1 scale. Claude Haiku 4.5 -- Lower cost, lower quality. Grant writing quality is the core value proposition; the cost saving does not justify the quality tradeoff.
Recommendation:	<i>Amazon Bedrock eu-west-2 is the only production-grade path to running Claude in UK/EEA region today. Azure AI Foundry is worth monitoring as it matures but offers no advantage that justifies migration before or shortly after launch.</i>

New Addition to Consider

Axiom -- Structured Log Management	Consider adding -- P5.4
---	--------------------------------

Axiom provides a searchable, chartable log management layer with a native Vercel log drain integration. Once the drain is configured -- a single URL added in Vercel project settings -- every function log line becomes queryable in real time, including the structured latency logs already emitted by all three AI routes.

This directly closes the remaining gap from GAP-27: AI route latency data is being logged in structured form but currently has nowhere to land that supports alerting or trend analysis. Sentry handles exceptions; Axiom handles telemetry. They are complementary, not competing.

Cost: Free tier -- 10 GB/month, 90-day retention. Zero cost at v1 scale.

Setup effort: ~15 minutes. One log drain URL in Vercel project settings; no SDK changes required.

What it replaces: Nothing -- complements Sentry. Does not replace error tracking.

When to add: P5.4 -- alongside Sentry performance monitoring configuration.

Data region: Axiom EU region available; select at account setup to satisfy C13.

Summary Matrix

Ref	Component	Current selection	Verdict	Note
T01	Framework	Next.js App Router + TypeScript	Retain	No credible alternative at this scale and team size
T02	Database	Supabase PostgreSQL -- London	Retain	Unique bundling of Postgres + Auth + RLS + Storage in UK region
T03	Authentication	Supabase Auth (bundled)	Retain	Native RLS integration scopes every DB query at the database level automatically
T04	Hosting	Vercel Pro -- London (1hr1)	Retain	90 s timeout requirement eliminates Cloudflare -- the only serious cost-competitive alternative
T05	Email	Resend (free tier)	Retain -- Monitor	AWS SES is the natural future migration -- same AWS footprint, fraction of cost per email
T06	Error tracking	Sentry EU	Retain	Best Next.js three-runtime coverage; EU data region satisfies C13
T07	Rate limiting	Upstash Redis -- sliding window	Retain	Better burst protection than a DB counter at zero additional cost
T08	CI pipeline	GitHub Actions -- 4 jobs	Retain	Native to GitHub, zero configuration overhead, free for public repos
T09	Tests	Vitest	Retain	Playwright for E2E is the natural post-launch complement

AI	AI provider/model	Bedrock eu-west-2 -- Sonnet 4.6	Retain -- Monitor	Monitor Azure AI Foundry (Claude in UK South) as it matures
--	Log management	Not in stack	Add -- P5.4	Axiom -- ~15 min setup, closes AI latency gap, EU region, free tier